AI5707 – Pattern Recognition and Machine Learning

Assignment 7: GMM Models and Pattern Recognition

Due: Oct 26 2026

1 Objective

In this assignment, you will learn to build GMMs for each class using GMM models:

$$p(\mathbf{x}) = \Sigma_{i=1}^{K} \pi_i \mathcal{N}(\mathbf{x}; \mu_\mathbf{i}, \mathbf{C}_i)$$

2 Problem definition

1. Using the data generated in Assignment 6, build GMM models for each of the classes using 80% of the data, and perform classification. Create the confusion matrix.
2. Use the MNIST dataset available on the website. The MNIST dataset consists of images of hand written characters corresponding to digits 0-9. Each group will work with 3 different digits. Treat each digit as an image. Each normalised image is made up of 784 pixel values. There are about 60000 images in the training set. Use the 3 digits assigned to your team. You could further compress the 784 pixel values to represent them as image patches. Treat each image or image patch as training data. Build a GMM model for each digit using ONLY the training data.
3. Using the model trained as above, perform the classification on the test data. Record your results.
4. When digits are confused, you must try to explain it with the GMM model trained – primarily look at the boundaries between classes.